

hardware, etc.—can play in academia and how academia should use this effectively to nurture these movements in return, for a win-win relationship.

We will first discuss the basic driving force of a common mindset, and the associated implications. Then we look at specific aspects such as content, standards, software/hardware, etc. Though licensing is a major issue in this regard, we will ignore the licence discussion in this article, it being more appropriate for someone with a legal background to talk about this.

Philosophical match

Academia is about sharing knowledge—to enrich the giver and the receiver. One builds on the knowledge created by others, and shares the enriched knowledge with others to let it grow further. In a broad sense, software and content can be seen as embodiments of knowledge, differing, perhaps, in the way the knowledge is captured and represented. And hence, the notion that software needs to be available with the full source code is something natural to the academic community.

It was the commercial companies that introduced the concept of hiding the source behind a compilation process, to 'protect' what they call as their 'intellectual property', in turn to ensure that copies (and even more so, any modifications) are not done by the customers. The matching mindsets have many implications for both academia and the open movement.

The popularity of open source is the highest in academia. The pricing issue is certainly a major factor here, since academia often has the severest budget constraints in acquiring resources such as software and equipment. The ability for academia to understand the philosophy behind the open movement, in the sense of sharing knowledge freely, also plays a major role. Their contribution in pushing the quality and quantity of open source higher has been significant.

Researchers, even earlier, used to contribute the software developed for their research work, often in cutting areas of development, to public use. The Moodle learning management system, LaTeX document processing system, etc. are good examples of high-quality systems coming out this way.

Another implication of this link is the growth of open standards. FOSS is strongly based on the community metaphor of the *bazaar* development model described by Eric S. Raymond (ESR), which brings together a number of people from around the world to work on a common system. The roles for each are open, and how much they contribute to the final system is also open. There are only internal deadlines set by oneself.

This necessarily demands explicit efforts to reduce the learning curve for others, and transparency in shared data structures. Open standards, where the complete description is freely accessible to everyone, and where the standards evolve from collective contributions, becomes a natural choice. Not surprisingly, most of the open source programs use open standards wherever available. Here, formats and conventions invented for one system are reused for another, if relevant.

This brings us to the idea of building on what is existing—another typically academic mindset. As Newton remarked, "If I have seen farther, it is by standing on the shoulders of giants." Research literature that builds on earlier results and acknowledges them by citation, and new software programs that reuse and extend existing software programs are clearly echoing the same mindset.

Starting from scratch every time, does not take you very far, in trying out new ideas in an already rich landscape. Starting from something that provides a close approximation to what you are looking for, and extending it, as appropriate, is more productive. As ESR has remarked: "A good programmer knows what code to write; a great programmer knows what to rewrite." This kind of reuse and extension necessitates the openness associated with the open movements. In fact, all these characteristics are fundamental to the growth and sustenance of open movements. And the outcome of these movements, in turn, contributes substantially to the growth and effectiveness of academia.

Open content

The last few years have seen tremendous growth in open content, where the content is declared free for use, just as was done for FOSS. Comparable to FOSS licences like GPL, LGPL, etc. a group of licences also evolved to provide legal sanctity to this move. These are known as Creative Commons and offer a family of licences embodying the core idea of openness, and providing options for permission to modify, retaining attribution, and so on.

Perhaps, the most classic example of open content today is Wikipedia, which takes the openness to the extreme, allowing anyone with an account (that can be freely obtained) to modify any of the pages. However, the quality of the content on Wikipedia is generally very good, and some formal studies have also shown this empirically. Though there are topics involving strong controversies, where one often sees a series of continuous modifications by the opposing sides to support their stand.

For most academic content, the Wikipedia offers excellent reference/learning material with additional links, images, and so on. Lacking even a core group to filter modifications as is done with open source software, the high-quality of content indeed shows the feasibility of the approach.

The EU-funded SELF project exploits such resources, to even form course material for university courses, dynamically. Other examples of open content are the 'million books' of Project Gutenberg, the audio book collection at LibriVox, the number of video repositories of Google video, YouTube, and so on. The movement got its momentum from the MIT initiative of open source ware, which has, in turn, led to a wider initiative of the open knowledge initiative (OKI) involving a number of partner institutions to share such resources.

It may be noted that these different set-ups follow different norms as far as their policy of use and modifications are concerned. A lot of open source software documentation and learning material are also available in such free content. These